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Disturbance observer based actor-critic learning control for uncertain nonlinear systems

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Abstract This paper investigates the disturbance observer based actor-critic learning control for a class of uncertain nonlinear systems in the presence of unmodeled dynamics and time-varying disturbances. The proposed control algorithm integrates a filter-based design method with actor-critic learning architecture and disturbance observer to circumvent the unmodeled dynamic and the time-varying disturbance. To be specific, the actor network is employed to estimate the unknown system dynamic, the critic network is developed to evaluate the control performance, and the disturbance observer is leveraged to provide efficient estimation of the compounded disturbance which includes the time-varying disturbance and the actor-critic network approximation error. Consequently, high-gain feedback is avoided and the improved tracking performance can be expected. Moreover, a composite weight adaptation law for actor network is constructed by utilizing two types of signals, the cost function and the modeling error. Eventually, theoretical analysis demonstrates that the developed controller can guarantee bounded stability. Extensive simulations and experiments on a robot manipulator are implemented to validate the performance of the resulted control strategy.

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1. Introduction

Owing to its significance, both from a practical and a theoretical perspective, the control design of uncertain nonlinear systems has been a major research topic over the past decades.[1–4] Some remarkable control approaches can be found in Refs. 5–7, including backstepping control, adaptive control, and observer-based nonlinear control, to name a few. Among them, adaptive control is an effective approach to address unknown parameters, and hereafter, its combination with backstepping technique plays an important role in the control of nonlinear systems. However, all these aforementioned methods cannot be directly applied to the nonlinear systems containing completely unknown dynamic structure, which hinders their widespread application.

In recent years, scholars have observed that Neural Network (NN) displays excellent ability in dealing with unknown nonlinearity due to its universal function approximation properties, and substantial control problems have been addressed by utilizing the NN.[8–10] For instance, in Ref. 11, an adaptive neural tracking control problem was investigated for strict feedback nonlinear systems with unmodeled dynamics. By introducing robust control and disturbance observer techniques, the authors in Refs. 12–13 presented robust adaptive neural control and disturbance observer based adaptive neural

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control, which can handle both unmodeled dynamics and time-varying disturbances. Different from the traditional NN-based control, an advanced neural learning control strategy is further developed according to actor-critic learning architecture.[14,15] To be specific, the actor-critic learning architecture consists of two networks: an actor network and a critic network. The actor network is leveraged to approximate an unknown function and generate action or control signal, while the critic network is leveraged to evaluate the control performance in the actor-critic learning architecture. The actor-critic learning architecture with a generalized learning structure can be easily applied to control other nonlinear systems.

Enlightened by the philosophy in Ref. 15, extensive control approaches were studied by integrating the actor-critic learning architecture with traditional control approaches for unknown nonlinear systems. In Refs. 16–18, the actor-critic architecture has been successfully applied to estimate the unknown nonlinearity online and achieved satisfactory results, yet they ignore the negative influence of time-varying unknown disturbances on control performance. However, for the practical systems (e.g., vehicular systems, robot manipulators and unmanned aerial vehicles[19–21]), the time-varying perturbations always exist. The appearance of time-varying disturbance could generate some unexpected results, such as degrading the control performance and leading to system divergence, and it is difficult to circumvent the influence of the time-varying disturbances with actor-critic learning control alone. Moreover, the approximation inaccuracy caused by actor-critic learning also influences the tracking performance. In this regard, the above-mentioned factors facilitate the combination of the actor-critic control and robust control or disturbance observer based control. In Ref. 22, the actor-critic structure is used to estimate the modeling uncertainties that exist in a small unmanned helicopter, and a discontinuous sliding model based robust component is introduced to eliminate the influence of the approximation error of actor network and unknown disturbance. To overcome the discontinuities of control input, a prescribed performance fault-tolerant control approach is developed in Ref.23 by integrating the actor-critic learning scheme with a Robust Integral of the Sign of Error (RISE) feedback term, which requires less system information and achieves the asymptotic stability. However, large feedback gains are required to resist unknown disturbances in robust control, which will reduce the stability margin, even stimulate high-frequency dynamics, and then result in system instability. Inspired by the feedforward design, disturbance observer based control[24–27] can be used to estimate the impact of disturbance and further used for compensation. In Refs. 28,29, a reinforcement learning based controller integrated with disturbance observer is established to reject the real-time external disturbance, which can guarantee the robust stability and the nominal performance even for the uncertain plant, and obtain satisfactory results in the numerical simulation. However, both of them merely focus on the regulation control problem.

Inspired by the aforementioned challenges, a disturbance observer based actor-critic learning control is developed for a class of uncertain nonlinear systems in the presence of unknown dynamic and time-varying disturbance. To cope with the unknown dynamic, the actor-critic learning architecture is developed to provide the feedforward compensation by approximating unknown nonlinearity. Considering the effect of the time-varying disturbance and actor-critic network approximation error on tracking performance, the actor-critic learning algorithm is combined with the disturbance observer to circumvent the effects of the above factors. In addition, a composite weight adaptation law for actor network is constructed by utilizing two types of signals, the cost function and the modeling error. Consequently, high-gain feedback is avoided and the improved tracking performance can be achieved. Eventually, extensive simulations and experiments on a robot manipulator are implemented to validate the performance of the resulted control strategy.

The key contributions of this paper are listed as follows:

(1) An actor-critic learning architecture is developed to estimate the unknown system dynamic online, which requires less model information and improves the robustness to unmodeled dynamic effectively.
(2) A disturbance observer is effectively combined to compensate for the time-varying disturbance and actor-critic network approximation error, which avoids high-gain feedback and achieves improved tracking performance.

To the best of our knowledge, few studies have integrated actor-critic learning architecture with disturbance observer for tracking control of uncertain nonlinear systems with unmodeled dynamic and time-varying unknown disturbance.

The remainder of this paper is organized as follows. The problem description is provided in Section 2. Section 3 states the disturbance observer based actor-critic learning control scheme and system stability analysis. Simulation and experimental studies on a single-link robot are provided in Section 4, and conclusions are drawn in Section 5.

2. Problem description

Consider a class of the nth-order Multiple Input Multiple Output (MIMO) nonlinear systems with the following form:

$$\begin{cases} \dot{\boldsymbol{x}}_i = \boldsymbol{x}_{i+1} i = 1, 2, \cdots, n-1 \\ \dot{\boldsymbol{x}}_n = \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} + \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{d}(t) \\ \boldsymbol{y}(t) = \boldsymbol{x}_1 \end{cases} \tag{1}$$

where $\boldsymbol{x} = [\boldsymbol{x}_1^{\mathrm{T}}, \boldsymbol{x}_2^{\mathrm{T}}, \cdots, \boldsymbol{x}_n^{\mathrm{T}}]^{\mathrm{T}} \in \Re^{mn}$ denotes the system state vector with $\boldsymbol{x}_i \in \Re^m$, which is assumed to be available for measurement; $\boldsymbol{u} \in \Re^m$ is the control input; $\boldsymbol{y} \in \Re^m$ is the system output; $\boldsymbol{f} \in \Re^m$ is the unknown smooth nonlinear function, $\boldsymbol{g}(\boldsymbol{x}) \in \Re^m$ is the known nonzero function, and $\boldsymbol{d}(t) \in \Re^m$ is the time-varying disturbance.

The main control objective of this study is to propose a disturbance observer based actor-critic learning control strategy to achieve high tracking accuracy with unmodeled dynamic and time-varying unknown disturbance. To facilitate the presentation, some related assumptions and lemmas are necessary.

Assumption 1. The reference trajectory $\boldsymbol{y}_{\mathrm{r}}(t) \in \Re^m$ and its n-derivative $\boldsymbol{y}_{\mathrm{r}}^{(n)}$ are available, smooth, and bounded.

Assumption 2. The time-varying disturbance $\boldsymbol{d}(t)$ and its first derivative are bounded, which satisfy $\|\boldsymbol{d}\| \leqslant d_{\mathrm{m}}, \|\dot{\boldsymbol{d}}\| \leqslant \bar{d}_{\mathrm{m}}$.

Lemma 1. The NN universal approximation property indicates that a continuous function $\boldsymbol{\Phi} : S \to \Re^{N_1}$ (S is a compact set) can be approximated as

$$\boldsymbol{\Phi}(\boldsymbol{x}) = \boldsymbol{W}^{\mathrm{T}}\boldsymbol{\varphi}(\boldsymbol{x}) + \boldsymbol{\varepsilon}(\boldsymbol{x}) \tag{2}$$

where $\boldsymbol{x} \in \Re^{N_2}$ is the input vector, $\boldsymbol{W} \in \Re^{N_3 \times N_1}$ is the ideal weight matrix, and N_1, N_2 and N_3 are numbers of neurons in the output, input, and hidden layer, respectively. $\boldsymbol{\varphi}(\boldsymbol{x}) \in \Re^{N_3}$ is the nonlinear activation function. According to Ref. 22, the ideal NN weights, nonlinear activation function and approximation error are assumed to be bounded by $\|\boldsymbol{W}\| \leqslant W_{\mathrm{m}}, \|\boldsymbol{\varphi}\| \leqslant \varphi_{\mathrm{m}}, \|\dot{\boldsymbol{\varphi}}\| \leqslant \bar{\varphi}_{\mathrm{m}}, \|\boldsymbol{\varepsilon}\| \leqslant \varepsilon_{\mathrm{m}}, \|\dot{\boldsymbol{\varepsilon}}\| \leqslant \bar{\varepsilon}_{\mathrm{m}}$.

3. Main results

In this scenario, the disturbance observer based actor-critic learning control strategy for uncertain nonlinear system Eq. (1) will be presented. First, we provide the backstepping controller design based on a filter-based design approach. Then, we design the actor-critic network to deal with the unknown nonlinear dynamic, where the critic network is leveraged to evaluate the control performance while the actor network is leveraged to approximate the unknown function. The architecture of the developed control scheme is depicted as Fig. 1.

3.1. Controller design

To quantify the aforementioned control objective, the tracking error $\boldsymbol{z}_1 \in \Re^m$ is defined as $\boldsymbol{z}_1 = \boldsymbol{y} - \boldsymbol{y}_{\mathrm{r}}$, and the following filtered tracking errors are introduced to facilitate the controller design

$$\begin{cases} \boldsymbol{z}_2 = \dot{\boldsymbol{z}}_1 + k_1\boldsymbol{z}_1 \\ i = 3, 4, \cdots, n \\ \boldsymbol{z}_i = \dot{\boldsymbol{z}}_{i-1} + k_{i-1}\boldsymbol{z}_{i-1} \end{cases} \tag{3}$$

where $k_1, k_2, \cdots, k_{n-1} \in \Re$ denote positive control gains. By substituting Eq. (1) into Eq. (3), the dynamic of filtered tracking error $\boldsymbol{z}_n$ can be written as

$$\begin{aligned} \dot{\boldsymbol{z}}_n &= \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} + \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{d} - \boldsymbol{y}_{\mathrm{r}}^{(n)} + k_1\boldsymbol{z}_1^{(n-1)} \\ &\quad + k_2\boldsymbol{z}_2^{(n-2)} + \cdots + k_{n-1}\boldsymbol{z}_{n-1}^{(1)} \\ &= \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} + \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{d} - \boldsymbol{y}_{\mathrm{r}}^{(n)} + \bar{\boldsymbol{k}}\,\bar{\boldsymbol{z}} \end{aligned} \tag{4}$$

where $\bar{\boldsymbol{k}} = [k_1, k_2, \cdots, k_{n-1}]$, $\bar{\boldsymbol{z}} = [\boldsymbol{z}_1^{(n-1)}; \boldsymbol{z}_2^{(n-2)}; \cdots; \boldsymbol{z}_{n-1}^{(1)}]$, and the unknown nonlinear function $\boldsymbol{f}(\boldsymbol{x})$ can be approximated by an NN-based actor network

$$\boldsymbol{f}(\boldsymbol{x}_{\mathrm{a}}) = \boldsymbol{W}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}(\boldsymbol{x}_{\mathrm{a}}) + \boldsymbol{\varepsilon}_{\mathrm{a}} \tag{5}$$

where $\boldsymbol{x}_{\mathrm{a}} = [\boldsymbol{x}_1^{\mathrm{T}}, \boldsymbol{x}_2^{\mathrm{T}}, \cdots, \boldsymbol{x}_n^{\mathrm{T}}]^{\mathrm{T}}$ denotes the input vector, $\boldsymbol{W}_{\mathrm{a}}$ denotes the weight vector of actor network and $\boldsymbol{\varepsilon}_{\mathrm{a}}$ denotes the actor network function reconstruction inaccuracy, which satisfies $\|\boldsymbol{\varepsilon}_{\mathrm{a}}\| \leqslant \varepsilon_{\mathrm{am}}$ and $\|\dot{\boldsymbol{\varepsilon}}_{\mathrm{a}}\| \leqslant \bar{\varepsilon}_{\mathrm{am}}$. Defining $\boldsymbol{\zeta} = \boldsymbol{\varepsilon}_{\mathrm{a}} + \boldsymbol{d}$, the expression in Eq. (4) can be further expressed as

$$\dot{\boldsymbol{z}}_n = \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} + \hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} - \tilde{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} + \boldsymbol{\zeta} - \boldsymbol{y}_{\mathrm{r}}^{(n)} + \bar{\boldsymbol{k}}\bar{\boldsymbol{z}} \tag{6}$$

where $\tilde{\boldsymbol{W}}_{\mathrm{a}} = \hat{\boldsymbol{W}}_{\mathrm{a}} - \boldsymbol{W}_{\mathrm{a}}$ and $\hat{\boldsymbol{W}}_{\mathrm{a}}$ is the estimate of $\boldsymbol{W}_{\mathrm{a}}$, which will be introduced later.

Generally, time-varying external disturbances can be estimated by using model-based disturbance observers.[30] Herein, the time-varying external disturbance and the residual function reconstruction inaccuracy of the actor network are integrated as $\boldsymbol{\zeta}$. The adaptive neural disturbance observer for estimating the lumped disturbance $\boldsymbol{\zeta}$ is designed by using neural network approximation

$$\begin{cases} \dot{\boldsymbol{h}} = -\alpha(\boldsymbol{h} + \alpha\boldsymbol{x}_n) - \alpha(\hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} + \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u}) \\ \hat{\boldsymbol{\zeta}} = \boldsymbol{h} + \alpha\boldsymbol{x}_n \end{cases} \tag{7}$$

where $\boldsymbol{h} \in \Re^n$ is an internal state and $\alpha \in \Re$ is a positive constant. Therefore, the control input $\boldsymbol{u}$ can be given by

$$\boldsymbol{u} = \boldsymbol{g}^{-1}(\boldsymbol{x})(\boldsymbol{y}_{\mathrm{r}}^{(n)} - \bar{\boldsymbol{k}}\bar{\boldsymbol{z}} - k_n\boldsymbol{z}_{\mathrm{n}} - \hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} - \hat{\boldsymbol{\zeta}}) \tag{8}$$

where $k_n \in \Re$ denotes a positive control gain. Substituting Eq. (8) into Eq. (6), the dynamics of filtered error $\boldsymbol{z}_n$ can be rewritten as

$$\dot{\boldsymbol{z}}_n = -k_n\boldsymbol{z}_n - \tilde{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} + \tilde{\boldsymbol{\zeta}} \tag{9}$$

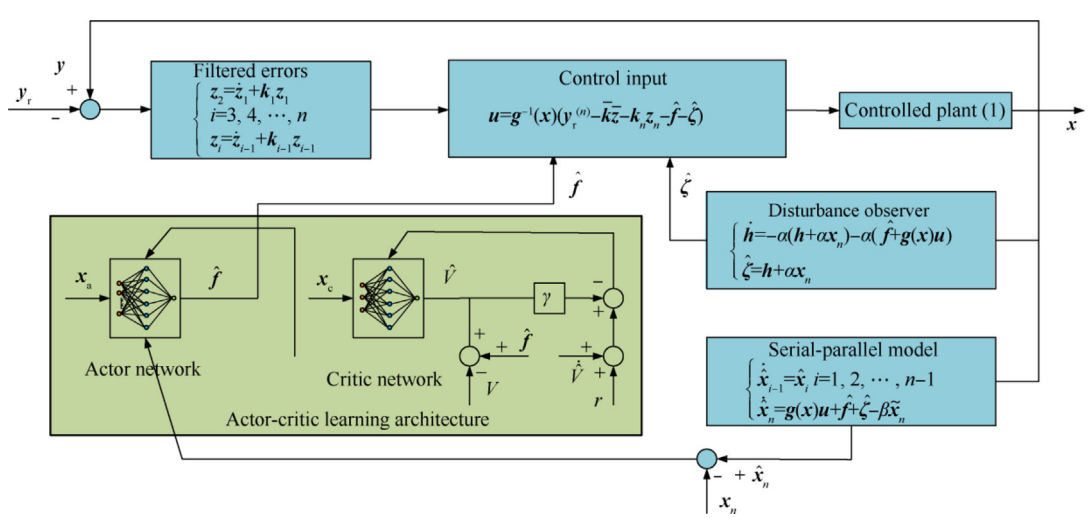

Fig. 1 Structure of disturbance observer based actor-critic learning control strategy.

with $\tilde{\zeta} = \zeta - \hat{\zeta}$ and

$$\dot{\tilde{\zeta}} = -\alpha\tilde{\zeta} + \alpha\tilde{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} + \dot{\zeta} \tag{10}$$

Remark 1. Apart from external disturbance, the residual function reconstruction inaccuracy of actor network is estimated by designing adaptive neural disturbance observer, which is different from conventional disturbance observers.[30] The performance and robustness of the control plant can be greatly improved by utilizing the adaptive neural disturbance observer for disturbance compensation.

3.2. Critic network design

The critic network is utilized to provide the evaluation function for the current strategy, which can test the performance of the current policy and generate rewards/punishments as the feedback for adaptive learning. Therefore, we introduce the infinite horizon performance index function as follows:

$$V(t) = \int_t^{\infty} \exp[-\gamma(s-t)]r(s)\mathrm{d}s \tag{11}$$

where $\gamma > 0$ represents a discount factor, which can guarantee the boundedness of the cost function even if the reference trajectory does not converge to zero, and $r(t)$ represents an instantaneous cost function

$$r(t) = \boldsymbol{z}_1^{\mathrm{T}}\boldsymbol{Q}\boldsymbol{z}_1 + \boldsymbol{u}^{\mathrm{T}}\boldsymbol{R}\boldsymbol{u} \tag{12}$$

where $\boldsymbol{Q} \in \Re^{m\times m}$ and $\boldsymbol{R} \in \Re^{m\times m}$ are the weighting matrices for the lumped tracking error $\boldsymbol{z}_1$ and control input $\boldsymbol{u}$, respectively. To achieve optimal control, the cost-to-go function is supposed to be minimized. Given that it is difficult to obtain the cost-to-go function, an NN-based critic network is introduced

$$V(\boldsymbol{x}_{\mathrm{c}}) = \boldsymbol{W}_{\mathrm{c}}^{\mathrm{T}}\boldsymbol{\varphi}(\boldsymbol{x}_{\mathrm{c}}) + \varepsilon_{\mathrm{c}} \tag{13}$$

where $\boldsymbol{x}_{\mathrm{c}} = \boldsymbol{z}_1$ denotes the input vector, $\boldsymbol{W}_{\mathrm{c}}$ denotes the weight vector of critic network and ε_{c} denotes the critic network function reconstruction inaccuracy, which satisfies $\|\boldsymbol{\varepsilon}_{\mathrm{c}}\| \leqslant \varepsilon_{\mathrm{cm}}$ and $\|\dot{\boldsymbol{\varepsilon}}_{\mathrm{c}}\| \leqslant \bar{\varepsilon}_{\mathrm{cm}}$. And the cost-to-go function can be approximated by

$$\hat{V}(\boldsymbol{x}_c) = \hat{\boldsymbol{W}}_{\mathrm{c}}^{\mathrm{T}}\boldsymbol{\varphi}(\boldsymbol{x}_c)$$

where $\hat{\boldsymbol{W}}_{\mathrm{c}}$ is the estimate of $\boldsymbol{W}_{\mathrm{c}}$.

The weight vector $\widehat{\boldsymbol{W}}_{\mathrm{c}}$ is selected to minimize the objection function $E_{\mathrm{c}} = 0.5e_{\mathrm{c}}^{\mathrm{T}}e_{\mathrm{c}}$, and according to Eq. (11) and Eq. (12), the prediction error $e_{\mathrm{c}}(\cdot)$ can be expressed as

$$e_{\mathrm{c}} = r(t) + \dot{\hat{V}}(t) - \gamma\hat{V}(t) \tag{15}$$

and the critic network weight parameters are updated by the following update law:

$$\begin{aligned}\dot{\widehat{\boldsymbol{W}}}_{\mathrm{c}} &= -k_{\mathrm{c1}}e_{\mathrm{c}}(-\gamma\boldsymbol{\varphi}_{\mathrm{c}} + \nabla\boldsymbol{\varphi}_{\mathrm{c}}\dot{\boldsymbol{x}}_{\mathrm{c}}) - k_{\mathrm{c2}}\widehat{\boldsymbol{W}}_{\mathrm{c}} \\ &= -k_{\mathrm{c1}}(r + \widehat{\boldsymbol{W}}_{\mathrm{c}}^{\mathrm{T}}\boldsymbol{\Lambda})\boldsymbol{\Lambda} - k_{\mathrm{c2}}\widehat{\boldsymbol{W}}_{\mathrm{c}}\end{aligned} \tag{16}$$

where $k_{\mathrm{c1}}, k_{\mathrm{c2}}$ are positive parameters and $\boldsymbol{\Lambda} = -\gamma\boldsymbol{\varphi}_{\mathrm{c}} + \nabla\boldsymbol{\varphi}_{\mathrm{c}}\dot{\boldsymbol{x}}_{\mathrm{c}}$.

3.3. Actor network design

The actor network is leveraged to estimate the unknown function $\boldsymbol{f}(\boldsymbol{x})$ that exists in Eq. (4), which can generate the appropriate control policy by gradually accumulating the system experience. The approximation of $\boldsymbol{f}(\boldsymbol{x})$ is designed as follows:

$$\hat{\boldsymbol{f}}(\boldsymbol{x}_{\mathrm{a}}) = \hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}(\boldsymbol{x}_{\mathrm{a}}) \tag{17}$$

Define a prediction error $\boldsymbol{e}_{\mathrm{a}}(\cdot)$ as

$$\boldsymbol{e}_{\mathrm{a}} = \boldsymbol{K}_{\mathrm{V}}(\hat{V} - V_{\mathrm{d}}) + \hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} \tag{18}$$

where $\boldsymbol{K}_{\mathrm{V}} \in \Re^{m}$ is the positive design parameter and $V_{\mathrm{d}} = 0$ is the ideal value of cost-to-go. The weight vector $\widehat{\boldsymbol{W}}_{\mathrm{a}}$ is selected to minimize the objection function $E_{\mathrm{a}} = 0.5\boldsymbol{e}_{\mathrm{a}}^{T}\boldsymbol{e}_{\mathrm{a}}$ and the actor network weight parameters are updated by the following update law:

$$\dot{\widehat{\boldsymbol{W}}}_{\mathrm{a}} = -k_{\mathrm{a1}}\boldsymbol{\varphi}_{\mathrm{a}}(\boldsymbol{K}_{\mathrm{V}}\hat{V} + \hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}}) - k_{\mathrm{a2}}\widehat{\boldsymbol{W}}_{\mathrm{a}} \tag{19}$$

with positive parameters k_{a1} and k_{a2}. To further improve the convergence of the estimated weights $\widehat{\boldsymbol{W}}_{\mathrm{a}}$ and the function approximation precision of actor network, another prediction error $\tilde{\boldsymbol{x}}_n$, named modeling error,[24] is defined as $\tilde{\boldsymbol{x}}_n = \hat{\boldsymbol{x}}_n - \boldsymbol{x}_n$, in which $\hat{\boldsymbol{x}}_n$ can be obtained by constructing the following serial-parallel estimation model [31]:

$$\begin{cases}\dot{\hat{\boldsymbol{x}}}_{i-1} = \hat{\boldsymbol{x}}_i i = 1, 2, \cdots, n-1 \\ \dot{\hat{\boldsymbol{x}}}_n = \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} + \hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} + \hat{\zeta} - \beta\tilde{\boldsymbol{x}}_n\end{cases} \tag{20}$$

and the dynamic equation $\tilde{\boldsymbol{x}}_n$ is written as

$$\dot{\tilde{\boldsymbol{x}}}_n = -\beta\tilde{\boldsymbol{x}}_n - \tilde{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}} - \tilde{\zeta} \tag{21}$$

in which $\beta \in \Re$ is a positive constant.

Therefore, the actor network weight parameters are adjusted by the following composite update law:

$$\dot{\widehat{\boldsymbol{W}}}_{\mathrm{a}} = -k_{\mathrm{a1}}\boldsymbol{\varphi}_{\mathrm{a}}(\boldsymbol{K}_{\mathrm{V}}\hat{V} + \hat{\boldsymbol{W}}_{\mathrm{a}}^{\mathrm{T}}\boldsymbol{\varphi}_{\mathrm{a}}) - k_{\mathrm{a2}}\widehat{\boldsymbol{W}}_{\mathrm{a}} - k_{\mathrm{a3}}\boldsymbol{\varphi}_{\mathrm{a}}\tilde{\boldsymbol{x}}_{\mathrm{n}}^{\mathrm{T}} \tag{22}$$

where k_{a3} is a positive parameter.

Remark 2. Different from the traditional action network weights updating,[18,22,23] a composite weight adaptation law for actor network is constructed by using both prediction error $\boldsymbol{e}_{\mathrm{a}}$ and modeling error $\tilde{\boldsymbol{x}}_n$, which ensures that the estimated weights $\widehat{\boldsymbol{W}}_{\mathrm{a}}$ converge better to unknown weights $\boldsymbol{W}_{\mathrm{a}}$ and the more precise approximation of the nonlinear function is achieved.[32]

3.4. Stability analysis

Theorem 1. In consideration of the nonlinear system Eq. (1) in the presence of unmodeled dynamic and time-varying disturbance, if the control input Eq. (8), the critic network weight adaptive law Eq. (16), the actor network weight adaptive law Eq. (22) and the adaptive neural disturbance observer Eq. (7) are designed, then all system signals are bounded. Proof details are given in Appendix A.

4. Simulation and experiments

4.1. Simulation

To substantiate the feasibility and effectiveness of the developed control strategy, we consider a two-degree-of-freedom robot manipulator (see Ref. 32) with the following dynamic equation:

$$\boldsymbol{M}(\boldsymbol{q})\ddot{\boldsymbol{q}} + \boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})\dot{\boldsymbol{q}} + \boldsymbol{F}(\dot{\boldsymbol{q}}) + \boldsymbol{\tau}_{\mathrm{d}} = \boldsymbol{\tau} \tag{23}$$

where $\boldsymbol{q},\dot{\boldsymbol{q}},\ddot{\boldsymbol{q}}$ denote the position, velocity, and acceleration, respectively, $\boldsymbol{M}(\boldsymbol{q})$ is the inertia matrix, $\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}})$ is the centripetal-Coriolis matrix, $\boldsymbol{F}(\dot{\boldsymbol{q}})$ is the friction, $\boldsymbol{\tau}_{\mathrm{d}}$ is the external disturbance, and $\boldsymbol{\tau}$ is the control input.

The matrix $\boldsymbol{M}(\boldsymbol{q}),\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}),\boldsymbol{F}(\dot{\boldsymbol{q}})$ and $\boldsymbol{\tau}_{\mathrm{d}}$ are given as follows:

$$\boldsymbol{M}(\boldsymbol{q}) = \begin{bmatrix} p_1 + 2p_3\cos(q_2) & p_2 + p_3\cos(q_2) \\ p_2 + p_3\cos(q_2) & p_2 \end{bmatrix}$$

$$\boldsymbol{C}(\boldsymbol{q},\dot{\boldsymbol{q}}) = \begin{bmatrix} -p_3\sin(q_2)\dot{q}_2 & -p_3\sin(q_2)(\dot{q}_1+\dot{q}_2) \\ p_3\sin(q_2)\dot{q}_1 & 0 \end{bmatrix}$$

$$\boldsymbol{F}(\dot{\boldsymbol{q}}) = \begin{bmatrix} f_{\mathrm{d1}} & 0 \\ 0 & f_{\mathrm{d2}} \end{bmatrix}\begin{bmatrix} \dot{q}_1 \\ \dot{q}_2 \end{bmatrix}, \boldsymbol{\tau}_{\mathrm{d}} = \begin{bmatrix} \tau_{\mathrm{d1}} \\ \tau_{\mathrm{d2}} \end{bmatrix}$$

where $p_1 = 3.473\ \mathrm{kg\cdot m^2}$, $p_2 = 0.196\ \mathrm{kg\cdot m^2}$, $p_3 = 0.242\ \mathrm{kg\cdot m^2}$, $f_{\mathrm{d1}} = 5.3\ \mathrm{N\cdot m\cdot s}$, $f_{\mathrm{d2}} = 1.1\ \mathrm{N\cdot m\cdot s}$, $\tau_{\mathrm{d1}} = 3\sin(t)$ and $\tau_{\mathrm{d2}} = 0.2\sin(t)$.

Then the dynamics in Eq. (23) can be transformed into the state-space equation considered in this paper, i.e.,

$$\begin{cases} \dot{\boldsymbol{x}}_1 = \boldsymbol{x}_2 \\ \dot{\boldsymbol{x}}_2 = \boldsymbol{g}(\boldsymbol{x})\boldsymbol{u} + \boldsymbol{f}(\boldsymbol{x}) + \boldsymbol{d}(t) \end{cases} \tag{24}$$

with $\boldsymbol{x}_1 = [q_1, q_2]^{\mathrm{T}}, \boldsymbol{x}_2 = [\dot{q}_1, \dot{q}_2]^{\mathrm{T}}, \boldsymbol{g}(\boldsymbol{x}) = \boldsymbol{M}^{-1}(\boldsymbol{x}_1)$, $\boldsymbol{u} = \boldsymbol{\tau}$, $\boldsymbol{f}(\boldsymbol{x}) = -\boldsymbol{M}^{-1}[\boldsymbol{C}(\boldsymbol{x}_1,\boldsymbol{x}_2)\boldsymbol{x}_2 + \boldsymbol{F}(\boldsymbol{x}_2)]$ and $\boldsymbol{d}(t) = -\boldsymbol{M}^{-1}\boldsymbol{\tau}_{\mathrm{d}}$.

The following two control strategies are compared to validate the effectiveness of the proposed approach:

Controller 1. This is the proposed controller or more specifically, actor-critic learning control integrated with disturbance observer. The control parameters are chosen as $k_1 = 30$, $k_2 = 10$, $\alpha = 20$, $\beta = 100$, $k_{\mathrm{c1}} = 2$, $k_{\mathrm{c2}} = 0.1$, $k_{\mathrm{a1}} = 20$, $k_{\mathrm{a2}} = 1$ and $k_{\mathrm{a3}} = 5$. The initial weights of actor-critic networks are chosen as $\widehat{\boldsymbol{W}}_{\mathrm{a}} = \mathrm{zeros}(10,4)$ and $\widehat{\boldsymbol{W}}_{\mathrm{c}} = \mathrm{zeros}(10,2)$. The discount factor is chosen as $\gamma = 0.1$, the positive matrices $\boldsymbol{Q}$ and $\boldsymbol{R}$ in the cost function are selected as $\boldsymbol{Q} = \mathrm{diag}([50\ 200])$ and $\boldsymbol{R} = \mathrm{diag}([0.1\ 0.1])$, respectively.

Controller 2. This is the actor-critic learning control approach without disturbance feedforward compensation. To ensure fair comparison, the selected control parameters are consistent with Controller 1.

The reference trajectories of two joints are chosen as $y_{\mathrm{r1}} = 0.6\sin(3.14t)\cdot[1-\exp(-t)]$ and $y_{\mathrm{r2}} = 0.8\sin(3.14t)\cdot[1-\exp(-t)]$. The simulation results are depicted as Figs. 2-6. As depicted in Fig. 2 and Fig. 3, the integrated Controller 1 controller can follow the reference signal well and achieve the best tracking performance in aspects of convergence speed and steady tracking error since the disturbance observer is introduced. In comparison of the last 20 s, the maximum amplitude of steady tracking error $\boldsymbol{M}_{z_1} = [0.0009, 0.0058]$ rad under Controller 1, while the maximum amplitude of steady tracking error $\boldsymbol{M}_{z_1} = [0.0021, 0.0103]$ rad under Controller 2.

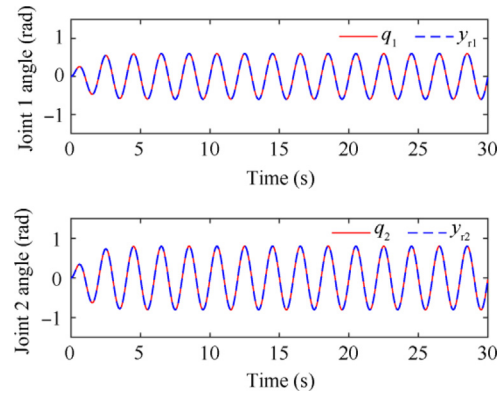

Fig. 2 Tracking performance with proposed controller Controller 1.

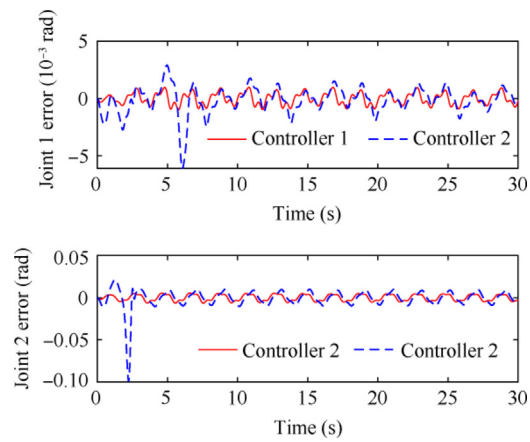

Fig. 3 Tracking errors for Joints 1 and 2 under Controller 1 and Controller 2.

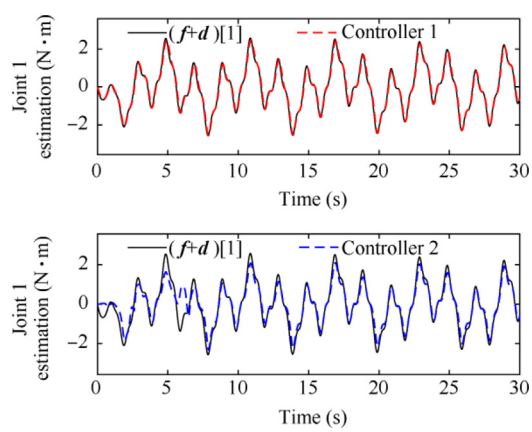

Fig. 4 Compound estimation $(\boldsymbol{f}+\boldsymbol{d})[1]$ for Joint 1 under Controller 1 and Controller 2.

The results in Figs. 4 and 5 depict the compound estimation of $\boldsymbol{f}+\boldsymbol{d}$, and it can be found that the composite estimation architecture established by actor-critic learning and disturbance observer can well approximate the unmodeled dynamic and time-varying disturbance in comparison of the estimation architecture only with actor-critic learning. This phenomenon

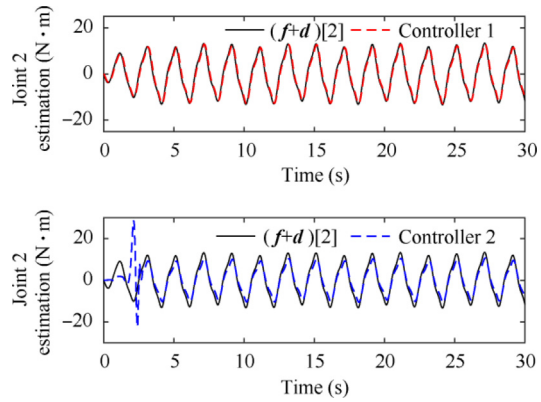

Fig. 5 Compound estimation $(\boldsymbol{f}+\boldsymbol{d})[2]$ for Joint 2 under Controller 1 and Controller 2.

explains that the accurate feedforward compensation can result in higher tracking accuracy. Eventually, the control inputs are shown in Fig. 6, which are regular and bounded.

4.2. Experiments

To further substantiate the priority of the developed control strategy, an experiment was conducted on a single-degree-of-freedom robot manipulator platform shown in Fig. 7. The test rig includes a bench case, a motor actuator (consisting of a DC motor Kollmorgen DH063A, an electrical driver Kollmorgen ServoStar 620, a rotary encoder Heidenhain ERN180, and a revolute joint), a link, a payload and a control module. The control module consists of a real-time control software composed of an Advantech PCI-1723 and a Heidenhain IK-220 counter card, and a monitoring software. The sampling time is 0.5 ms.

The dynamic of single-degree-of-freedom robot manipulator can also be written as the state-space form shown in Eq. (24) with $x_1 = q, x_2 = \dot{q}, x = [x_1, x_2]^{\mathrm{T}}$, $g(x) = J^{-1}$, $u = \tau$, $f(x) = -J^{-1}[F(\dot{q}) + G(q)]$ and $d(t) = -J^{-1}\tau_{\mathrm{d}}$, where $q, \dot{q}$ denote the position and velocity, respectively, $F(\dot{q})$ is the unknown

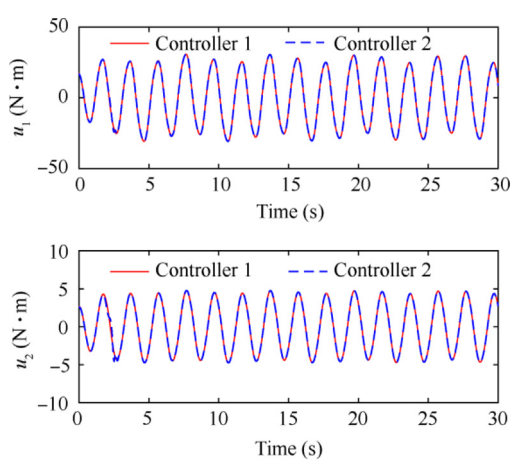

Fig. 6 Control inputs of two joints under Controller 1 and Controller 2.

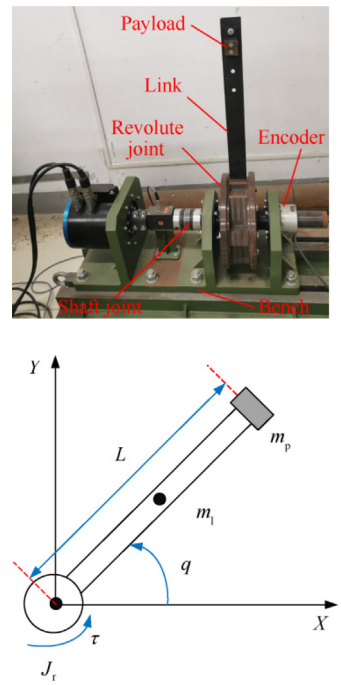

Fig. 7 A single-degree-of-freedom robot manipulator platform.

friction, $G(q)$ is the unknown gravity, τ_{d} is the external disturbance, τ is the control input, and $J = J_{\mathrm{r}} + J_{\mathrm{l}} + J_{\mathrm{p}}$ is the total moment of inertia with the joint moment of inertia J_{r}, the link moment of inertia $J_{\mathrm{l}} = m_{\mathrm{l}}L^2/3$ and the payload moment of inertia $J_{\mathrm{p}} = m_{\mathrm{p}}L^2$. The system parameters are provided as $J_{\mathrm{r}} = 0.3\ \mathrm{kg}\cdot\mathrm{m}^2$, $L = 0.5$ m, $m_{\mathrm{l}} = 0.5$ kg and $m_{\mathrm{p}} = 0 - 1$ kg.

Likewise, the aforementioned two controllers are still utilized to test in the experiments. The reference signal is chosen as $y_{\mathrm{r}} = 10[1 - \cos(3.14t)] \cdot [1 - \exp(-t)]$, the control parameters are chosen as $k_1 = 150$, $k_2 = 50$, $\alpha = 10$, $\beta = 50$, $k_{\mathrm{c1}} = 2$, $k_{\mathrm{c2}} = 0.5$, $k_{\mathrm{a1}} = 20$, $k_{\mathrm{a2}} = 2$ and $k_{\mathrm{a3}} = 5$. The initial weights of actor-critic networks are chosen as $\widehat{\boldsymbol{W}}_{\mathrm{a}} = \mathrm{zeros}(10, 2)$, $\hat{\boldsymbol{W}}_{\mathrm{c}} = \mathrm{zeros}(10, 1)$. The discount factor is chosen as $\gamma = 0.1$, and the positive matrices $\boldsymbol{Q}$ and $\boldsymbol{R}$ in the cost function are selected as $\boldsymbol{Q} = [50]$ and $\boldsymbol{R} = [1]$, respectively. In this scenario, the control performance with different payloads $m_{\mathrm{p}} = 0$ kg, $m_{\mathrm{p}} = 0.5$ kg and $m_{\mathrm{p}} = 1$ kg is tested, respectively. Furthermore, to quantitatively evaluate the tracking performance of the aforementioned controllers, three performance indices (i.e., the maximum M_z, average μ, and standard deviation σ) in Ref. 33 are introduced.

Case 1. The compared two controllers are tested under no-load condition with $m_{\mathrm{p}} = 0$ kg. The tracking errors and the performance indices (during the last 20 s) of the two controllers are presented in Fig. 8. From these results, it can be observed that the tracking performance of Controller 1 is improved compared with that of Controller 2 since the disturbance observer is integrated with actor-critic learning control, and the result of compound estimation $\hat{f} + \hat{d}$ acquired by action network and disturbance observer is depicted in Fig. 9.

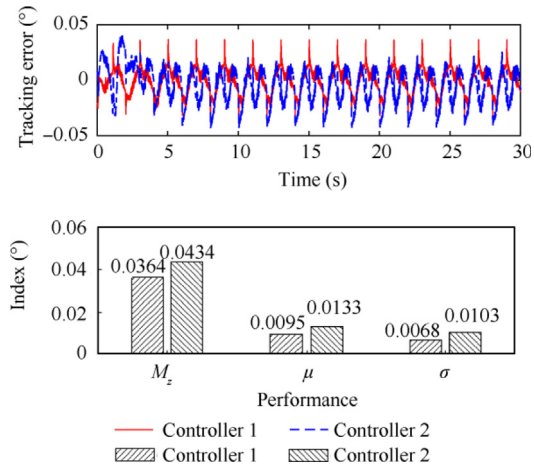

Fig. 8 System tracking errors of Controller 1 and Controller 2 under no-load condition with $m_p = 0$ kg.

Case 2. The compared two controllers are tested under light load condition with $m_p = 0.5$ kg. The tracking errors and the performance indices are shown in Fig. 10. It can be seen that the performance of Controller 1 still outperforms that of Controller 2. In comparison with the no-load condition, M_z of Controller 1 changed very little, and only increased by 1.6%, while that of Controller 2 increased by about two times, and this phenomenon illustrates that the actor-critic learning control alone is not enough to compensate for the impact of load change on the system. In addition, the result of compound estimation is depicted in Fig. 11. Compared with Fig. 9, it can be observed that as the payload becomes larger, the proposed composite estimation scheme can well adapt to the change of system uncertainty, which further illustrates that Controller 1 can effectively compensate for the influence of load change.

Case 3. The compared two controllers are tested under heavy load condition with $m_p = 1$ kg. The tracking errors and the performance indices are shown in Fig. 12. Likewise, the performance of Controller 1 still outperforms that of Controller 2, and compared with the no-load condition, M_z of Controller 1 increased by 5.5%, while that of Controller 2 increased by about 2.3 times, which further illustrates that Controller 1 can effectively compensate for the impact of load change on the system. The result of compound estimation is depicted in Fig. 13. Compared with Fig. 9, it can be observed that as the payload becomes larger, Controller 1 can still well adapt to the change of system uncertainty.

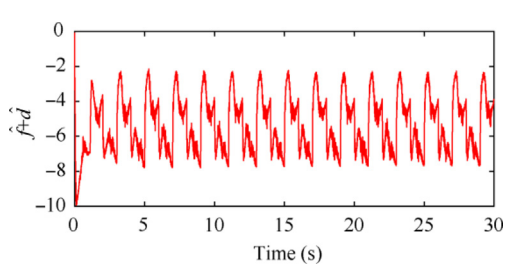

Fig. 9 Compound estimation of $\hat{f}+\hat{d}$ of Controller 1 under no-load condition with $m_p = 0$ kg.

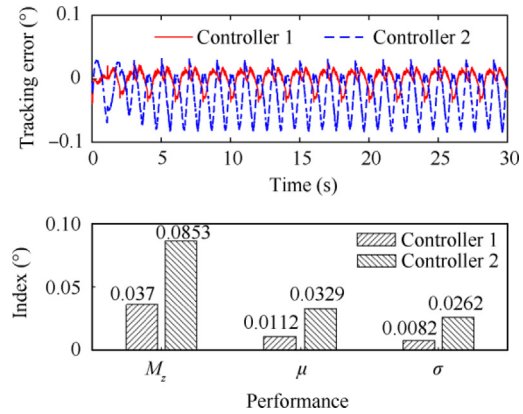

Fig. 10 System tracking errors of Controller 1 and Controller 2 under light load condition with $m_p = 0.5$ kg.

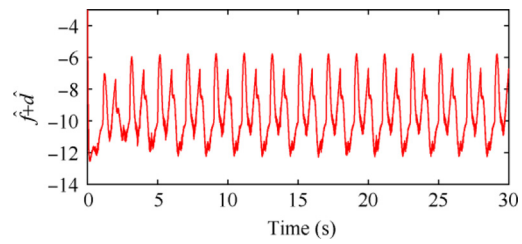

Fig. 11 Compound estimation of $\hat{f}+\hat{d}$ of Controller 1 under light load condition with $m_p = 0.5$ kg.

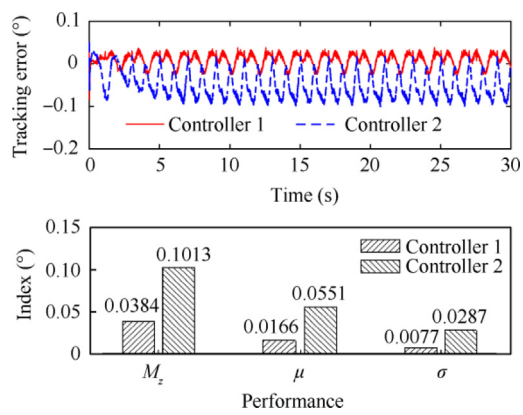

Fig. 12 System tracking errors of Controller 1 and Controller 2 under heavy load condition with $m_p = 1$ kg.

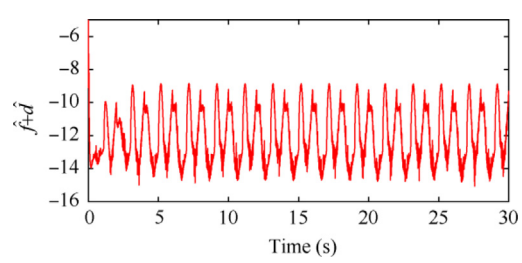

Fig. 13 Compound estimation of $\hat{f}+\hat{d}$ of Controller 1 under heavy load condition with $m_p = 1$ kg.

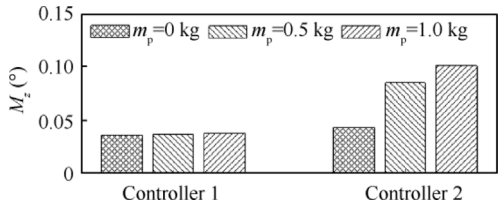

Fig. 14 M_z of two controllers under different payloads.

In order to more intuitively observe the impact of load change on the tracking performance of controllers Controller 1 and Controller 2, M_z of the two controllers under different payloads are collected together, as shown in Fig. 14. It is evident that Controller 1 can effectively compensate for the impact of load change on the system and maintain its tracking performance, while the tracking performance of Controller 2 gradually deteriorates with the increase of payload. To this end, the proposed controller Controller 1 is robust to unknown uncertainties and the improved tracking performance can be achieved.

5. Conclusions

In this paper, the disturbance observer based actor-critic learning control is investigated for a class of nonlinear systems in the presence of unknown dynamic and time-varying disturbance. A composite weight adaptation law for actor network is constructed by both cost function and modeling error, and a disturbance observer component is combined to compensate for the residual functional reconstruction inaccuracy caused by the actor network and the time-varying disturbance. Extensive simulations and experiments on a robot manipulator present that using the developed disturbance observer based actor-critic learning control strategy can effectively circumvent the influence of unknown system dynamic and time-varying disturbance, and higher tracking accuracy can be achieved. Considering that the full state information is required in the implementation of the developed control strategy, we will explore an output feedback control approach for uncertain nonlinear systems in the future.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

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Appendix A. Proof of Theorem 1. Consider the following Lyapunov candidate function:

$$L = L_1 + L_2 + L_3 + L_4 + L_5 \tag{A1}$$

where

$$L_1 = \sum_{i=1}^{n} \frac{1}{2} z_i^T z_i,\ L_2 = \frac{1}{2}\tilde{\zeta}^{\mathrm{T}}\tilde{\zeta},\ L_3 = \frac{1}{2}\mathrm{tr}(\tilde{W}_{\mathrm{c}}^{\mathrm{T}}\tilde{W}_{\mathrm{c}}),$$
$$L_4 = \frac{1}{2}\mathrm{tr}(\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\tilde{W}_{\mathrm{a}}),\ L_5 = \frac{1}{2}\tilde{x}_{\mathrm{n}}^{\mathrm{T}}\tilde{x}_{\mathrm{n}} \tag{A2}$$

Using Eq. (3) and Eq. (9), the derivative of L_1 can be expressed as

$$\begin{aligned}\dot{L}_1 &= -k_1 z_1^{\mathrm{T}} z_1 + z_1^{\mathrm{T}} z_2 - k_2 z_2^{\mathrm{T}} z_2 + z_2^{\mathrm{T}} z_3 + \cdots - k_{n-1} z_{n-1}^{\mathrm{T}} z_{n-1} \\ &\quad + z_{n-1}^{\mathrm{T}} z_n - k_n z_n^{\mathrm{T}} z_n + z_n^{\mathrm{T}}(-\tilde{W}_{\mathrm{a}}^{T}\varphi_{\mathrm{a}} + \tilde{\zeta}) \\ &\le -k_1\|z_1\|^2 + \frac{1}{2}\|z_1\|^2 + \frac{1}{2}\|z_2\|^2 - k_2\|z_2\|^2 \\ &\quad + \frac{1}{2}\|z_2\|^2 + \frac{1}{2}\|z_3\|^2 + \cdots - k_{n-1}\|z_{n-1}\|^2 \\ &\quad + \frac{1}{2}\|z_{n-1}\|^2 + \frac{1}{2}\|z_n\|^2 - k_n\|z_n\|^2 + \|z_n\|^2 \\ &\quad + \frac{1}{2}\|\tilde{\zeta}\|^2 + \frac{1}{2}\|\tilde{W}_{\mathrm{a}}\|^2\|\varphi_{\mathrm{a}}\|^2\end{aligned} \tag{A3}$$

Using Eq. (10), the derivative of L_2 can be expressed as

$$\begin{aligned}\dot{L}_2 &= -\alpha\tilde{\zeta}^{\mathrm{T}}\tilde{\zeta} + \alpha\tilde{\zeta}^{\mathrm{T}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}} + \tilde{\zeta}^{\mathrm{T}}\dot{\zeta} \\ &\leqslant -\alpha\|\tilde{\zeta}\|^2 + \tfrac{1}{2}\alpha\|\tilde{\zeta}\|^2 + \tfrac{1}{2}\alpha\|\tilde{W}_{\mathrm{a}}\|^2\|\varphi_{\mathrm{a}}\|^2 \\ &\qquad + \tfrac{1}{2}\|\tilde{\zeta}\|^2 + \tfrac{1}{2}\|\dot{\zeta}\|^2 \\ &\leqslant -\tfrac{1}{2}(\alpha-1)\|\tilde{\zeta}\|^2 + \tfrac{1}{2}\alpha\|\tilde{W}_{\mathrm{a}}\|^2\|\varphi_{\mathrm{a}}\|^2 + \tfrac{1}{2}\|\dot{\zeta}\|^2\end{aligned} \tag{A4}$$

Using Eq. (13) and Eq. (16), the derivative of L_3 can be expressed as.

$$\begin{aligned}\dot{L}_3 &= -k_{\mathrm{c1}}\tilde{W}_{\mathrm{c}}^{\mathrm{T}}(r(t) + \widehat{W}_{\mathrm{c}}^{\mathrm{T}}\Lambda)\Lambda - k_{c2}\tilde{W}_{\mathrm{c}}^{\mathrm{T}}\widehat{W}_{\mathrm{c}} \\ &= -k_{\mathrm{c1}}\tilde{W}_{\mathrm{c}}^{\mathrm{T}}(-\widehat{W}_{\mathrm{c}}^{\mathrm{T}}\Lambda + \varepsilon_{\mathrm{c}} + \widehat{W}_{\mathrm{c}}^{\mathrm{T}}\Lambda)\Lambda - k_{\mathrm{c2}}\tilde{W}_{\mathrm{c}}^{\mathrm{T}}\left(W_{\mathrm{c}} + \tilde{W}_{\mathrm{c}}\right) \\ &\leqslant -\tfrac{1}{2}\left(k_{\mathrm{c1}}\lambda_{\min}\left(\Lambda\Lambda^{\mathrm{T}}\right) + k_{\mathrm{c2}}\right)\tilde{W}_{\mathrm{c}}^{\mathrm{T}}\tilde{W}_{c} + \tfrac{k_{\mathrm{c1}}}{2}\in_{\mathrm{c}}^{\mathrm{T}}\in_{\mathrm{c}} + \tfrac{k_{\mathrm{c2}}}{2}W_{\mathrm{c}}^{\mathrm{T}}W_{\mathrm{c}}\end{aligned} \tag{A5}$$

where $\boldsymbol{\varepsilon}_{\mathrm{c}} = \gamma\boldsymbol{\varepsilon}_{\mathrm{c}} - \dot{\boldsymbol{\varepsilon}}_{\mathrm{c}}$, which is bounded, i.e., $\|\boldsymbol{\varepsilon}_{\mathrm{c}}\| \leqslant \varepsilon_{\mathrm{cm}}$.

Using Eq. (22), the derivative of L_4 can be expressed as

$$\begin{aligned}\dot{L}_4 &= -k_{\mathrm{a1}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}}(K_{\mathrm{V}}\widehat{W}_{\mathrm{c}}^{\mathrm{T}}\varphi_{\mathrm{c}} + \widehat{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}}) - k_{\mathrm{a2}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\widehat{W}_{\mathrm{a}} \\ &\qquad - k_{\mathrm{a3}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi\tilde{x}_n^{\mathrm{T}} \\ &= -k_{\mathrm{a1}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}} - k_{\mathrm{a1}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}}(W_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}} + K_{\mathrm{V}}\widehat{W}_{\mathrm{c}}^{\mathrm{T}}\varphi_{\mathrm{c}}) \\ &\qquad - k_{\mathrm{a2}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}(W_{\mathrm{a}} + \tilde{W}_{\mathrm{a}}) - k_{\mathrm{a3}}\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}}\tilde{x}_n^{\mathrm{T}} \\ &\leqslant -\tfrac{1}{2}(k_{\mathrm{a1}}\lambda_{\min}(\varphi_{\mathrm{a}}\varphi_{\mathrm{a}}^{\mathrm{T}}) + k_{\mathrm{a2}} - k_{\mathrm{a3}}\|\varphi_{\mathrm{a}}\|^2)\tilde{W}_{\mathrm{a}}^{\mathrm{T}}\tilde{W}_{\mathrm{a}} \\ &\qquad + k_{\mathrm{a1}}\|W_{\mathrm{a}}\|^2\|\varphi_{\mathrm{a}}\|^2 + 2k_{\mathrm{a1}}K_{\mathrm{V}}^{\mathrm{T}}K_{\mathrm{V}}\|\varphi_{\mathrm{c}}\|^2\tilde{W}_{\mathrm{c}}^{\mathrm{T}}\tilde{W}_{\mathrm{c}} \\ &\qquad + 2k_{\mathrm{a1}}K_{\mathrm{V}}^{\mathrm{T}}K_{\mathrm{V}}\|W_{\mathrm{c}}\|^2\|\varphi_{\mathrm{c}}\|^2 + \tfrac{k_{\mathrm{a2}}}{2}W_{\mathrm{a}}^{\mathrm{T}}W_{\mathrm{a}} + \tfrac{k_{\mathrm{a3}}}{2}\|\tilde{x}_n\|^2\end{aligned} \tag{A6}$$

Using Eq. (21), the derivative of L_5 can be expressed as

$$\dot{L}_5 = \tilde{x}_n^{\mathrm{T}}(-\beta\tilde{x}_n - \tilde{W}_{\mathrm{a}}^{\mathrm{T}}\varphi_{\mathrm{a}} - \tilde{\zeta}) \le -(\beta-1)\|\tilde{x}_n\|^2 + \frac{1}{2}\|\tilde{\zeta}\|^2 + \frac{1}{2}\|\tilde{W}_{\mathrm{a}}\|^2\|\varphi_{\mathrm{a}}\|^2 \tag{A7}$$

Then combining Eqs. (A3)–(A7), the first derivative of L can be written as

$$\begin{aligned}\dot{L} &\leqslant -(k_1 - 0.5)\|z_1\|^2 - (k_i - 1)\|z_i\|^2 - \cdots - (k_n - 1.5)\|z_n\|^2 \\ &\quad -\tfrac{1}{2}(k_{\mathrm{c1}}\lambda_{\min}(\Lambda\Lambda^{\mathrm{T}}) + k_{\mathrm{c2}} - 4k_{\mathrm{a1}}K_{\mathrm{V}}^{\mathrm{T}}K_{\mathrm{V}}\|\varphi_{\mathrm{c}}\|^2)\|\tilde{W}_{\mathrm{c}}\|^2 \\ &\quad -\tfrac{1}{2}(k_{\mathrm{a1}}\lambda_{\min}(\varphi_{\mathrm{a}}\varphi_{\mathrm{a}}^{\mathrm{T}}) - (k_{\mathrm{a3}} + \alpha + 2)\|\varphi_{\mathrm{a}}\|^2 + k_{\mathrm{a2}})\|\tilde{W}_{\mathrm{a}}\|^2 \\ &\quad -\tfrac{1}{2}(\alpha - 3)\|\tilde{\zeta}\|^2 - \tfrac{1}{2}(2\beta - k_{\mathrm{a3}} - 2)\|\tilde{x}_n\|^2 + \rho_1 \\ &\leqslant -\rho_0 L + \rho_1\end{aligned} \tag{A8}$$

where

$$\begin{aligned}\rho_0 = \min\{&2(k_1-0.5), 2(k_2-1), \cdots, 2(k_n-1.5), \alpha-3, 2\beta \\ &-k_{a3}-2, k_{c1}\lambda_{\min}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\mathrm{T}}) + k_{c2} - 4k_{a1}\boldsymbol{K}_{\mathrm{V}}^{\mathrm{T}}\boldsymbol{K}_{\mathrm{V}}\|\boldsymbol{\varphi}_{\mathrm{c}}\|^2, \\ &k_{a1}\lambda_{\min}(\boldsymbol{\varphi}_{\mathrm{a}}\boldsymbol{\varphi}_{\mathrm{a}}^{\mathrm{T}}) - (k_{a3}+\alpha+2)\|\boldsymbol{\varphi}_{\mathrm{a}}\|^2 + k_{a2}\} \\ \rho_1 = 0.5(&k_{c1}\epsilon_{\mathrm{cm}}^2 + \bar{\varepsilon}_{\mathrm{am}}^2 + \bar{d}_{\mathrm{m}}^2) + \varepsilon_{\mathrm{am}}^2 + d_{\mathrm{m}}^2 + (2k_{a1}\boldsymbol{K}_{\mathrm{V}}^{\mathrm{T}}\boldsymbol{K}_{\mathrm{V}}\|\boldsymbol{\varphi}_{\mathrm{cm}}\|^2 \\ &+0.5k_{c2})\|\boldsymbol{W}_{\mathrm{cm}}\|^2 + (0.5k_{a2} + k_{a1}\|\boldsymbol{\varphi}_{\mathrm{am}}\|^2)\|\boldsymbol{W}_{\mathrm{am}}\|^2\end{aligned}$$

To ensure $\rho_0 > 0$, the following conditions must be fulfilled:

$$\begin{aligned}&k_1 > 0.5, k_2 > 1, \cdots, k_n > 1.5 \\ &\alpha > 3, 2\beta - k_{a3} - 2 > 0 \\ &k_{c1}\lambda_{min}(\boldsymbol{\Lambda}\boldsymbol{\Lambda}^{\mathrm{T}}) + k_{c2} - 4k_{a1}\boldsymbol{K}_{\mathrm{V}}^{\mathrm{T}}\boldsymbol{K}_{\mathrm{V}}\|\boldsymbol{\varphi}_{\mathrm{c}}\|^2 > 0 \\ &k_{a1}\lambda_{min}(\boldsymbol{\varphi}_{\mathrm{a}}\boldsymbol{\varphi}_{\mathrm{a}}^{\mathrm{T}}) + k_{a2} - (k_{a3}+\alpha+2)\|\boldsymbol{\varphi}_{\mathrm{a}}\|^2 > 0\end{aligned} \tag{A9}$$

Solving the aforementioned differential inequality Eq. (A8), we have

$$L(t) \leqslant (L(0) - \frac{\rho_1}{\rho_0})\exp(-\rho_0 t) + \frac{\rho_1}{\rho_0} \leqslant L(0) + \frac{\rho_1}{\rho_0} \tag{A100}$$

Consequently, all system signals are bounded according to the definition of L in Eq. (A1).

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